

Resilience-driven restoration model for interdependent infrastructure networks

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ABSTRACT

Critical infrastructure networks such as electric power, water distribution, natural gas, transportation and telecommunications are the backbone of modern societies as they provide them with the services that are essential for their continuous functioning. However, these infrastructure networks are not isolated from each other but instead, most of them rely on one another to be functional. Hence, they are highly vulnerable to any disruptive event (e.g., components deteriorating, terrorist attacks, or natural disasters) which makes their restoration more challenging task for decision makers. In this paper, we study the restoration problem of a system of interdependent infrastructure networks following a disruption event considering different disruptions scenarios. We propose a resilience-driven multi-objective restoration model using mixed-integer programming that aims to maximize the resilience of the system of interdependent infrastructure networks while minimizing the total cost associated with the restoration process. The restoration model considers the availability of limited time and resources and provides a prioritized list of components, nodes or links, to be restored along with assigning and scheduling them to the available work crews. The proposed model is illustrated through a generated system of interdependent power-water networks, however it is applicable to any physically interdependent networks.

1. Introduction

A *critical infrastructure network* is defined as a network of independent, mostly privately-owned, man-made systems and processes that function collaboratively and synergistically to produce and distribute a continuous flow of essential goods and services [59]. Hence, critical infrastructure networks such as electric power, water distribution, natural gas, transportation and telecommunications are the backbone of modern societies as they provide them with the services that are essential for their continuous functioning.

Where previous work in planning for disruptions to critical infrastructure networks emphasized prevention and protection, such planning has recently shifted more broadly to capture the ability of infrastructure networks to withstand a disruption and recovery timely from it. The ability to withstand, adapt to, and recover from a disruption is generally referred to as *resilience* [8], a definition with which many would largely agree [3, 4, 26]. For critical infrastructures in particular, the Infrastructure Security Partnership [58] described that a resilient infrastructure sector would “prepare for, prevent, protect against, respond or mitigate any anticipated or unexpected significant threat or

event” and “rapidly recover and reconstitute critical assets, operations, and services with minimum damage and disruption.”

The study of critical infrastructure networks under disruption has matured considerably in the past fifteen years, where emphasis has been given to identifying topological descriptors [e.g., 6, 30, 44, 45] and flow-based descriptors [e.g., 37, 46, 48, 54] that enable the study of critical network components that lead to network vulnerability. On the other hand, other studies provide methods and algorithms for restoring critical infrastructure networks following the occurrence of a disruptive event by determining the set of disrupted network components that need to be restored to maximize the performance of the network, assigning these components to work crews, determining the restoration sequence of these components, or an integrated approach [e.g., 1, 21, 32, 33, 36, 42, 47, 60, 66].

However, these infrastructure networks rely on each other in different ways in order for them to be functional; hence they are increasingly becoming more interdependent (i.e., two infrastructure networks are said to be interdependent if there is a bidirectional relationship between them through which the state of each infrastructure is dependent on the other one) [53]. Hence, the US federal planning

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Notation

K	Set of infrastructure networks.
T	Set of available time periods.
Ψ	Set of interdependent nodes.
R^k	Set of available work crews for network $k \in K$.
N^k	Set of nodes in network $k \in K$.
N_s^k	Set of source nodes in network $k \in K$.
N_d^k	Set of demand nodes in network $k \in K$.
N'^k	Set of disrupted nodes in network $k \in K$.
L^k	Set of links in network $k \in K$.
L'^k	Set of disrupted links in network $k \in K$.
b_i^k	Amount of supply at node $i \in N_s^k$ in network $k \in K$.
u_{ij}^k	Capacity of link $(i, j) \in L^k$ in network $k \in K$.
μ^k	Weight of network $k \in K$.
S_o^k	Total slacks at all demand nodes in network $k \in K$ before a disruption.
S_d^k	Total slacks at all demand nodes in network $k \in K$ after a disruption.
fn_{it}^k	Cost of restoration for node $i \in N'^k$ in network $k \in K$ at time period $t \in T$.
fl_{ijt}^k	Cost of restoration for link $(i, j) \in L'^k$ in network $k \in K$ at time period $t \in T$.
p_{it}^k	Cost of unmet demand in node $i \in N_d^k$ in network $k \in K$ at

time period $t \in T$.

c_{ij}^k	Cost of flow through link $(i, j) \in L^k$ in network $k \in K$.
dn_i^k	Restoration duration of node $i \in N'^k$ in network $k \in K$.
dl_{ij}^k	Restoration duration of link $(i, j) \in L'^k$ in network $k \in K$.
s_{it}^k	Variable representing the amount of unmet demand at node $i \in N_d^k$ in network $k \in K$ at time period $t \in T$.
x_{ijt}^k	Variable representing the flow through link $(i, j) \in L^k$ in network $k \in K$ at time period $t \in T$.
z_i^k	Variable indicating whether or not node $i \in N'^k$ in network $k \in K$ is to be restored.
y_{ij}^k	Variable indicating whether or not link $(i, j) \in L'^k$ in network $k \in K$ is to be restored.
β_{it}^k	Variable indicating whether or not node $i \in N^k$ in network $k \in K$ is operational at time period $t \in T$.
α_{ijt}^k	Variable indicating whether or not link $(i, j) \in L^k$ in network $k \in K$ is operational at time period $t \in T$.
γ_{it}^{kr}	Variable indicating whether or not node $i \in N'^k$ in network $k \in K$ is restored by work crew $r \in R^k$ at time period $t \in T$.
δ_{ijt}^{kr}	Variable indicating whether or not link $(i, j) \in L'^k$ in network $k \in K$ is restored by work crew $r \in R^k$ at time period $t \in T$.

documents suggest (as do many across the globe) the importance in addressing critical infrastructure resilience in such a way that reflects its “interconnectedness and interdependency” [63]. Rinaldi et al. [53] stated in a seminal paper on infrastructure interdependencies that “critical infrastructures are highly interconnected and mutually dependent in complex ways, both physically and through a host of information and communications technologies.” Despite acknowledging the interdependent nature of “lifeline” infrastructures at roughly the same period [2], literature on the study of *interdependent* networks has only recently began appearing [e.g., 12, 13, 15, 18, 29, 39, 49, 50, 64, 67].

Rinaldi et al. [53] defined the interdependency between two infrastructure networks as “a bidirectional relationship between two infrastructures through which the state of each infrastructure influences or is correlated to the state of the other”. They classified the interdependencies between infrastructure networks into four categories: physical, cyber, geographical and logical interdependencies. The physical interdependency means that an output from an infrastructure network is an input to another one and vice versa. The cyber interdependency is existed for an infrastructure networks if it depends on information transmitted through an information infrastructure. Two infrastructure networks are geographically interdependent if they are affected by the same local disruptive event). Finally, all other types of interdependencies are classified as logical interdependencies.

As a result of the interdependencies between infrastructure networks, a disruption in some components of one of these infrastructure networks could lead to disruptions in other components of other dependent infrastructure networks. Therefore, they are highly vulnerable to any disruptive event (e.g., random failure, malevolent attack, natural disaster) that might propagate from one infrastructure network to another due to their interdependencies leading to inoperability in some or all of them [13, 18, 20, 41, 49, 61, 64]. Therefore, the restoration of such interdependent infrastructure networks following a disruptive event becomes more challenging for decision makers.

The problem of the restoration of a system of interdependent infrastructure networks after a disruptive event has been addressed with different approaches in the literature. Lee et al. [38] proposed an interdependent layer network model using mixed-integer programming (MIP) whose objective is to minimize the flow costs along with the slack

costs but not including the cost associated with the restoration process of the disrupted components. Moreover, it focuses on determining the set of disrupted components to be restored, though it does not specify the time they need to be restored at or the work crews that they are assigned to. On the other hand, Gong et al. [24] focused on scheduling the set of disrupted components assuming that they are known with a predefined due date for them. They provided a multi-objective restoration planning model and solve it through a benders decomposition approach. The model aims to minimize the cost, tardiness, and make-span to find the optimal restoration schedule for disrupted components. Coffrin et al. [17] studied the problem of restoring interdependent power and gas networks. They integrated two network-specific flow models (i.e., a linearized direct current flow model for the power network and a maximum flow model for the gas network) using MIP with the objective of maximizing the weighted sum of interdependent demand during over the restoration time horizon. However, the proposed model did not consider different restoration time for the disrupted component in addition for being developed for specific types of infrastructure networks. Cavdaroglu et al. [15] integrated the two approaches by Lee et al. [38] and Gong et al. [24] by providing a MIP model that integrates determining the set of disrupted components to be restored along with assigning and scheduling them to work crews. The objective of this model is to minimize the sum of the flow cost, unsatisfied demand cost, and installation and assignment cost associated with the full restoration of the interdependent infrastructure networks. González et al. [25] formulated an MIP model for the interdependent network design problem considering functionally and geographically interdependencies. The model determines the set of disrupted components to be restored and the order of their restoration with the objective of minimizing the cost associated with preparing geographical locations, restoration, unbalance from disconnection, and flow. However, the model does not specify which work crews should restore particular disrupted components. Also, they assumed that a work crew can restore any disrupted component from any infrastructure network without taking into account the skills required for each infrastructure network.

In this paper, we study the problem of the restoration of a system of interdependent infrastructure networks after the occurrence of a disruptive event considering different disruption scenarios. We propose a

resilience-driven multi-objective optimization model using mixed-integer programming (MIP) with the objectives of (i) maximizing the resilience of the system of interdependent infrastructure networks, and (ii) minimizing the costs associated with the restoration process, including flow, disruption, and restoration costs. Moreover, the proposed MIP restoration model takes into account the availability of the time and resources considering that there is a set of available resources or work crews or that are specific to each network (i.e., different than the assumption considered for the available models in the literature where a work crew can be assigned to restore a disrupted component in any network in the system of interdependent networks). It provides a set of prioritized restoration tasks to which to allocate and schedule available work crews considering the physical interdependence between the infrastructure networks such that the resilience of the system of interdependent infrastructure networks is maximized while the restoration cost is minimized. While the cost is generally a typical objective of restoration models in the literature, our proposed model aims to restore disrupted components of the system of interdependent networks to enhance its resilience (i.e., retain its performance prior to the occurrence of a disruptive event) while minimizing the cost associated to it (flow, restoration, and disruption costs). Hence, the proposed restoration model focuses on enhancing the resilience of the system of interdependent infrastructure networks to reach to a specific level with the minimum cost associated to it. Consequently, the disrupted components in the system of interdependent networks might not be all restored with the schedule resulting from this optimization problem, especially if they do not have any influence on the resilience of the system. While this work addresses interdependent network restoration, a resilience measure in the objective function will enable future explorations of the balance between “withstanding a disruption” and “recovering from a disruption,” which represent the two primary dimensions of resilience.

The remainder of the paper is organized as follows. Section 2 provides brief definitions and notation, an overview of network resilience, and disruption scenarios. The proposed resilience-driven multi-objective optimization model for the restoration of a system of interdependent infrastructure network is presented in Section 3. In Section 4, an illustrative example is presented with a generated system of interdependent power-water networks considering different disruptions scenarios. Finally, concluding remarks are provided in Section 5.

2. Methodological background

In this section, we discuss the background required to develop our proposed restoration model for a system of interdependent infrastructure networks.

2.1. Definitions and notation

In this work, we consider a network that is classified as an undirected graph and is denoted by $G = (N, L)$ where N is a set of n nodes and $L \subset \{(i, j): i, j \in N, i \neq j\}$ is a set of undirected links. The flow and capacity on link $(i, j) \in L$ are denoted by x_{ij} and u_{ij} , respectively. Let nodes s and t represent source and demand nodes, respectively. Nodes s and t are connected by a finite directed path, P , through a set of internal nodes in N and one or more links in L . The *maximum capacity of a path* equals the minimum capacity of all the links within that path (i.e., $\min_{(i,j) \in P} u_{ij}$) [22].

The objective of the $s - t$ max flow problem is to find the maximum flow from node s to node t by utilizing a subset of all possible paths between them accounting for link capacity. It can be formulated as a linear programming (LP) model as shown in Eqs. (1)–(3) [11]. The objective of the model, Eq. (1), is to maximize the flow from node s to node t , f_{st} , where $s, t \in N$ and $s \neq t$. Eq. (2) represents the flow conservation constraints, which ensure that the flow into and out of internal nodes are equal and the flow out of node s and into node t equal the maximum flow between them, f_{st} . Eq. (3) represents the non-negativity of the flow within the network along with the capacity constraint of the links to ensure that they do not exceed their capacities.

$$\max f_{st} \quad (1)$$

$$\text{s. t. } \sum_{(i,j) \in L} x_{ij} - \sum_{(j,i) \in L} x_{ji} = \begin{cases} f_{st} & \text{if } i = s \\ 0 & \text{if } i \in N \setminus \{s, t\} \\ -f_{st} & \text{if } i = t \end{cases} \quad (2)$$

$$0 \leq x_{ij} \leq u_{ij}, \quad i, j \in N \quad (3)$$

In this paper, we consider infrastructure networks with multiple source and demand nodes, which represent reality in many infrastructure networks (e.g., electric power networks) [55]. The multiple source, multiple demand network can be reduced to a single source, single demand network by using the Ford-Fulkerson [23] algorithm as follows:

- (i) Add a new source node s^* (i.e., super-source), and connect it to all source nodes by adding a link (s^*, s) from s^* to every source node s .
- (ii) Add a new demand node t^* (i.e., super-demand), and connect it to all demand nodes by adding a link (t, t^*) from every node t to t^* .
- (iii) Assign a capacity to each link (s^*, s) equal to the capacity of node s .
- (iv) Assign a capacity to each link (t, t^*) equal to the demand of node t .

The maximum flow from the super-source node, s^* , to the super-demand node, t^* , represents the total flow that can be supplied from the source nodes to the demand nodes within the network.

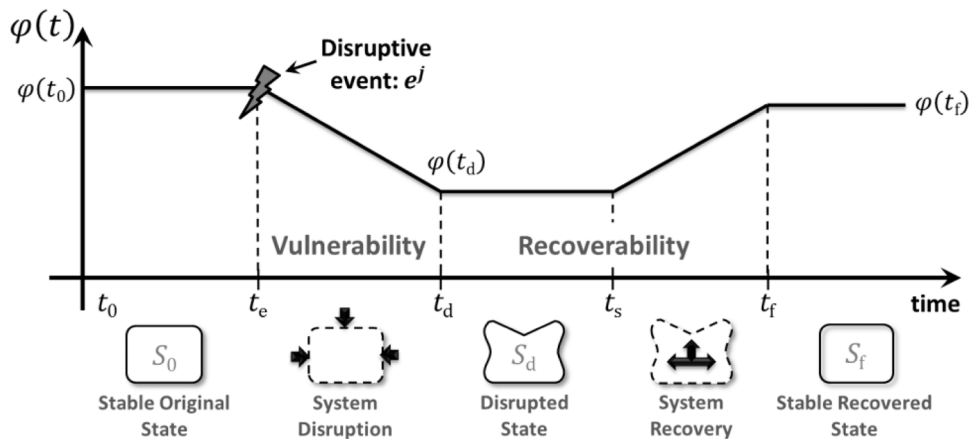


Fig. 1. Network performance, $\phi(t)$, across state transitions before, during, and after the occurrence of a disruptive event, e^j .

2.2. Network resilience

Several approaches have recently been offered in the literature to quantify the resilience [31] such as: the normalized shaded area underneath the performance function curve of a system [16], topological measures [57], the ratio of the probability of failure and recovery [40], among others. In this paper, we will consider the resilience paradigm based on network performance shown in Fig. 1 [7, 9, 10, 28, 51, 52]. Two primary dimensions of resilience, vulnerability and recoverability, are illustrated in Fig. 1. The vulnerability of a network can be defined as the magnitude of the damage to the network caused by a disruptive event [35], while the recoverability of a network refers to the speed at which the network recovers to a desired level of performance following a disruptive event [56].

Fig. 1 shows the performance of a network across different states over time, measured by function $\varphi(t)$, which describes the behavior of the network before, during and after the occurrence of a disruptive event, e^j . Accordingly, network resilience, $\mathcal{R}$, can be defined as the time dependent ratio of network recovery over its loss (i.e., $\mathcal{R}(t) = \text{Recovery}(t)/\text{Loss}(t)$ [28]) and can be mathematically represented by Eq. (4).

$$B_{\varphi}(t|e^j) = \frac{\varphi(t|e^j) - \varphi(t_d|e^j)}{\varphi(t_0) - \varphi(t_d|e^j)}, t \in (t_s, t_f) \quad (4)$$

The value of the network resilience, $\mathcal{R}_{\varphi}(t|e^j)$, at time t given the occurrence of a disruptive event, e^j , is between 0 and 1 where $\mathcal{R}_{\varphi}(t|e^j) = 1$ indicates that the network is fully resilient. In this work, we consider the maximum flow of an interdependent infrastructure network from its multiple supply nodes to its multiple demand nodes to be the function by which the network performance is measured, and its resilience is determined accordingly.

2.3. Disruption scenarios

Interdependent infrastructure networks are subjected to different scenarios of disruptions which could affect their performances differently. These disruption scenarios can be categorized into three groups [62]: random failures, malevolent attacks, and spatial failures. Random failures include common failures and manmade accidents such as aging, operating errors, and poor maintenance. For this scenario of disruptions, interdependent network components (nodes or links) are removed randomly with equal failure probability for all components. Malevolent attacks reflect intelligent attacks such as terrorism where important network components are targeted. Two scenarios are considered for malevolent attacks: capacity-based, where components with higher capacity are targeted, and degree-based, where components with higher degree (i.e., connections with other components) are targeted. For the capacity-based scenario, the capacities of the internal nodes are determined as the min of the sum of capacities for the incoming and outgoing links, respectively (i.e., $u_i^k = \min \{ \sum_{(j,i) \in L^k} u_{ji}^k, \sum_{(i,l) \in L^k} u_{il}^k \}$, where u_i^k is the capacity of node i in network k and L^k is the set of links in network k). For the degree-based scenario, the degree of link (i, j) is defined as the average of the degree of node i and node j (i.e., $\text{deg}_{ij} = \frac{1}{2}(\text{deg}_i + \text{deg}_j)$, where the degree of node i is the number of connections it has with other nodes in the network). Hence, interdependent network components are removed from their networks according to their capacity or degree where the components with the highest capacity or degree have higher failure probabilities than others. Finally, spatial failures capture natural disasters, such as earthquakes and hurricanes, that disrupt geographical locations. Consequently, the spatial disruptions affect the components of the interdependent networks that are spatially closed to each other (i.e., can be affected by the same local disruption). In this work, the area of the interdependent networks is divided into multiple regions where if a disruption occurs in a region, all the interdependent networks components within that region will be disrupted and hence removed from their networks.

3. Interdependent infrastructure network restoration

In this paper, we propose a multi-objective resilience-driven restoration optimization model using mixed-integer programming, aiming to maximize the resilience of the collective set of networks while minimizing the costs associated with the restoration process. It includes three sets of constraints: flow constraints for each infrastructure network, interdependencies constraints between networks, and assignment and scheduling constraints for the restoration tasks.

3.1. Assumptions

There are several assumptions and considerations for the proposed optimization model for the restoration of a system of interdependent infrastructure network:

- Each infrastructure network consists of a set of components (used generally to refer to nodes and/or links) that are subjected to disruptions (e.g., manmade accidents, malevolent attacks, or natural disasters). However, cascading disruptions are not considered here and are considered future work.
- Components of each infrastructure network are either completely disrupted or undisrupted following a disruptive event.
- Each disrupted component in each infrastructure network can be restored with different restoration durations (i.e., recovery durations are not fixed or the same for all disrupted components).
- A disrupted component is not operational unless it is completely restored. That is, the proposed model does not consider partial recovery where a component functions partially.
- Each supply node, demand node, and link in each infrastructure network has a known supply capacity, demand, and flow capacity, respectively.
- The flow cost through each link in each infrastructure network is known and fixed.
- The physical interdependence among different infrastructure networks is considered. That is, for a dependent node in an infrastructure network to be operational, it requires a specific node or nodes from another infrastructure network to also be operational.
- The number of available work crews for each infrastructure network (i.e., infrastructure-specific resources) for the restoration of its disrupted components is known and could be different from one infrastructure network to another.
- Each work crew in each infrastructure network can work on restoring a single disrupted component at a time.
- A work crew is not allowed to move from a disrupted component to another unless they complete the restoration of the previous one (i.e., the model considers a non-preemptive recovery process).

3.2. Notation

We are given a set of infrastructure networks, K , and a set of available time periods $T = \{1, \dots, \tau\}$. For each network $k \in K$, there is a set of nodes, N^k , and a set of links, L^k . Also, there is a set of source nodes, $N_s^k \subseteq N^k$, and a set of demand nodes, $N_d^k \subseteq N^k$. The disrupted components in each network $k \in K$ are denoted by N^k and L^k for disrupted nodes and disrupted links, respectively.

Let b_i^k be the amount of supply at node $i \in N_s^k$ in network $k \in K$ that is considered in this work to be the maximum flow from node $i \in N_s^k$ to all demand nodes in network $k \in K$. As such, b_i^k , assumed to be time independent, can be obtained by solving the model (1)–(3). The amount of unmet demand, called slack, at node $i \in N_d^k$ in network $k \in K$ at time period $t \in T$ is denoted by s_{it}^k . Hence, the total slacks at all demand nodes in network $k \in K$ after recovery at time period $t \in T$ equals $\sum_{i \in N_d^k} s_{it}^k$. We assume that resilience is a function of s_{it}^k , representing the extent to which demand in the network is not being met (as opposed to

using b_i^k , which is a fixed desired performance level of the interdependent infrastructure networks for our proposed model).

Accordingly, the slacks in the model represent the loss in the maximum flow, and reducing them to a desired level represents a means to measure the effectiveness of the restoration process. Hence, the resilience of the system of interdependent infrastructure networks, one of the main objectives of the restoration process, can be represented mathematically by Eq. (5), where μ^k is the weight of network $k \in K$ such that $\sum_{k \in K} \mu^k = 1$, and S_o^k and S_d^k represent the total slacks at all demand nodes in network $k \in K$ before and after a disruption, respectively (i.e., S_o^k refers to the total original slacks at all demand nodes in network $k \in K$ at time t_o , and S_d^k refers to the total slacks at all demand nodes in network $k \in K$ at time t_d following a disruptive event, e^j , as shown in Fig. 1). Hence, $\sum_{t=1}^T [t(S_d^k - \sum_{i \in N_d^k} s_{it}^k) - (t-1)(S_d^k - \sum_{i \in N_d^k} s_{i(t-1)}^k)]$ represents the cumulative recovery of network $k \in K$ over the restoration time horizon, where the recovery of the network at time $t \in T$ is determined by $S_d^k - \sum_{i \in N_d^k} s_{it}^k$, while its total loss is represented by $(S_d^k - S_o^k)$.

$$\sum_{k \in K} \mu^k \left[\frac{\sum_{t=1}^T \left[t \left(S_d^k - \sum_{i \in N_d^k} s_{it}^k \right) - (t-1) \left(S_d^k - \sum_{i \in N_d^k} s_{i(t-1)}^k \right) \right]}{\tau(S_d^k - S_o^k)} \right] \quad (5)$$

Another important and conflicting objective for the restoration process is the total cost of the restoration itself. Let the restoration cost for node $i \in N^k$ and link $(i, j) \in L^k$ in network $k \in K$ at time period $t \in T$ be denoted by f_{it}^k and f_{ijt}^k , respectively. The flow unitary cost (i.e., the cost for each unit of flow) through link $(i, j) \in L^k$ in network $k \in K$ is represented by c_{ij}^k , and p_{it}^k denotes the unit cost of unmet demand at node $i \in N_d^k$ in network $k \in K$ at time period $t \in T$. Hence, the system cost, which includes restoration cost, flow cost, and disruption cost (i.e., unmet demand), can be represented mathematically by Eq. (6), where z_i^k is a binary decision variable that equals 1 if node $i \in N^k$ in network $k \in K$ is to be restored and 0 otherwise, y_{ij}^k is a binary decision variable that equals 1 if link $(i, j) \in L^k$ in network $k \in K$ is to be restored and 0 otherwise, and x_{ijt}^k is a non-negative decision variable that represents the flow through link $(i, j) \in L^k$ in network $k \in K$ at time period $t \in T$.

$$\sum_{k \in K} \sum_{t \in T} \left(\sum_{i \in N^k} f_{it}^k z_i^k + \sum_{(i,j) \in L^k} f_{ijt}^k y_{ij}^k + \sum_{(i,j) \in L^k} c_{ij}^k x_{ijt}^k + \sum_{i \in N_d^k} p_{it}^k s_{it}^k \right) \quad (6)$$

Let the restoration duration of node $i \in N^k$ and link $(i, j) \in L^k$ in network $k \in K$ be denoted by dn_i^k and dl_{ij}^k , respectively. Each link $(i, j) \in L^k$ in network $k \in K$ has a capacity of u_{ij}^k . The status of each component at time period $t \in T$ is represented by two binary decision variables: β_{it}^k equals 1 if node $i \in N^k$ in network $k \in K$ is operational at time period $t \in T$ and 0 otherwise, and similarly, α_{ijt}^k equals 1 if link $(i, j) \in L^k$ in network $k \in K$ is operational at time period $t \in T$ and 0 otherwise. For each network $k \in K$, there is a set of available work crews or resources, R^k , that are specific to network k (e.g., expertise or equipment necessary to restore network k). The binary decision variables denoted by γ_{it}^{kr} and δ_{ijt}^{kr} represent the scheduling variables of the model, where γ_{it}^{kr} equals 1 if node $i \in N^k$ in network $k \in K$ is restored by work crew $r \in R^k$ in time period $t \in T$ and 0 otherwise. Likewise, δ_{ijt}^{kr} equals 1 if link $(i, j) \in L^k$ in network $k \in K$ is restored by work crew $r \in R^k$ in time period $t \in T$ and 0 otherwise. Finally, the interdependence between the networks is captured by Ψ , where $((i, k), (\bar{i}, \bar{k})) \in \Psi$ denote that node $\bar{i} \in N^{\bar{k}}$ in network $\bar{k} \in K$ depends physically on node $i \in N^k$ in network $k \in K$ in order to be operational.

3.3. Mathematical model

The proposed mathematical model focuses on optimizing two main

objectives: (i) a resilience objective, and (ii) a cost objective. The resilience objective, Eq. (7), maximizes the resilience of the system of interdependent networks over the restoration time horizon. However, the cost objective, Eq. (8), minimizes the total cost associated with the restoration process as the sum of flow cost, restoration cost, and disruption cost (i.e., unmet demand cost).

$$\max \sum_{k \in K} \mu^k \left[\frac{\sum_{t=1}^T \left[t \left(S_d^k - \sum_{i \in N_d^k} s_{it}^k \right) - (t-1) \left(S_d^k - \sum_{i \in N_d^k} s_{i(t-1)}^k \right) \right]}{\tau(S_d^k - S_o^k)} \right] \quad (7)$$

$$\min \sum_{k \in K} \sum_{t \in T} \left(\sum_{i \in N^k} f_{it}^k z_i^k + \sum_{(i,j) \in L^k} f_{ijt}^k y_{ij}^k + \sum_{(i,j) \in L^k} c_{ij}^k x_{ijt}^k + \sum_{i \in N_d^k} p_{it}^k s_{it}^k \right) \quad (8)$$

The two objectives of the proposed optimization model are subject to several sets of constraints: (i) network flow constraints, (ii) restoration constraints, (iii) interdependence constraints, (iv) logical link constraints for the network flow with restoration, and (v) constraints governing the nature of the decision variables. All sets of constraints are explained and formulated in the following sections.

Network flow constraints are represented by constraints (9)–(12). For each infrastructure network, the flow conservation at each of its (i) supply nodes, $i \in N_s^k$, (ii) transshipment nodes, $i \in N^k \setminus \{N_s^k, N_d^k\}$, and (iii) demand nodes, $i \in N_d^k$ in network $k \in K$ at time $t \in T$ is represented by constraints (9), (10), and (11), respectively. Constraint (12) ensures that the flow through link $(i, j) \in L^k$ in network $k \in K$ at time $t \in T$ does not exceed its capacity.

$$\sum_{(i,j) \in L^k} x_{ijt}^k \leq b_i^k, \quad \forall i \in N_s^k, \quad k \in K, \quad t \in T \quad (9)$$

$$\sum_{(i,j) \in L^k} x_{ijt}^k - \sum_{(j,i) \in L^k} x_{jit}^k = 0, \quad \forall i \in N^k \setminus \{N_s^k, N_d^k\}, \quad k \in K, \quad t \in T \quad (10)$$

$$\sum_{(j,i) \in L^k} x_{jit}^k + s_{it}^k = b_i^k, \quad \forall i \in N_d^k, \quad k \in K, \quad t \in T \quad (11)$$

$$x_{ijt}^k - u_{ij}^k \leq 0, \quad \forall (i, j) \in L^k, \quad k \in K, \quad t \in T \quad (12)$$

Restoration constraints (i.e., assignment and scheduling constraints) are represented by constraints (13)–(21). Constraints (13) and (14) ensure that if node $i \in N^k$ or link $(i, j) \in L^k$, respectively in network $k \in K$ is to be restored, it is scheduled to be restored by work crew $r \in R^k$ at time $t \in T$. Work crew $r \in R^k$ in network $k \in K$ can work on the restoration of at most one disrupted network component, node $i \in N^k$ or link $(i, j) \in L^k$, during time period $t \in T$, as shown in constraint (15). Constraints (16) and (17) ensure that if node $i \in N^k$ or link $(i, j) \in L^k$, respectively in network $k \in K$ is operational at time $t \in T$, it is completed by work crew $r \in R^k$. Constraints (18) and (19) ensure that node $i \in N^k$ or link $(i, j) \in L^k$, respectively in network $k \in K$ cannot be operational prior to its restoration duration. Similarly, work crew $r \in R^k$ cannot complete the restoration of node $i \in N^k$ or link $(i, j) \in L^k$ in network $k \in K$ prior to its restoration duration, as shown in constraints (20) and (21), respectively.

$$z_i^k = \sum_{r \in R^k} \sum_{t \in T} \gamma_{it}^{kr}, \quad \forall i \in N^k, \quad k \in K \quad (13)$$

$$y_{ij}^k = \sum_{r \in R^k} \sum_{t \in T} \delta_{ijt}^{kr}, \quad \forall (i, j) \in L^k, \quad k \in K \quad (14)$$

$$\sum_{i \in N^k} \sum_{l=1}^{\min\{\tau, t+dn_i^k-1\}} \gamma_{il}^{kr} + \sum_{(i,j) \in L^k} \sum_{l=1}^{\min\{\tau, t+dl_{ij}^k-1\}} \delta_{ijl}^{kr} \leq 1, \quad \forall k \in K, \quad r \in R^k, \quad t \in T \quad (15)$$

$$\beta_{it}^k \leq \sum_{r \in R^k} \sum_{l=1}^t \gamma_{il}^{kr}, \quad \forall i \in N^k, k \in K, t \in T \quad (16)$$

$$\alpha_{ijt}^k \leq \sum_{r \in R^k} \sum_{l=1}^t \delta_{ijl}^{kr}, \quad \forall (i, j) \in L^k, k \in K, t \in T \quad (17)$$

$$\sum_{t=1}^{dn_{it}^k-1} \beta_{it}^k = 0, \quad \forall i \in N^k, k \in K \quad (18)$$

$$\sum_{t=1}^{dl_{ij}^k-1} \alpha_{ijt}^k = 0, \quad \forall (i, j) \in L^k, k \in K \quad (19)$$

$$\sum_{r \in R^k} \sum_{t=1}^{dn_{it}^k-1} \gamma_{it}^{kr} = 0, \quad \forall i \in N^k, k \in K \quad (20)$$

$$\sum_{r \in R^k} \sum_{t=1}^{dl_{ij}^k-1} \delta_{ijl}^{kr} = 0, \quad \forall (i, j) \in L^k, k \in K \quad (21)$$

The physical interdependence among different infrastructure networks is represented by constraint (22), which ensures that for a node $\bar{i} \in N^k$ in network $\bar{k} \in K$ to be operational at time period $t \in T$, node $i \in N^k$ in network $k \in K$ must be operational at time $t \in T$, where $((i, k), (\bar{i}, \bar{k})) \in \Psi$.

$$\beta_{it}^{\bar{k}} - \beta_{it}^k \leq 0, \quad \forall ((i, k), (\bar{i}, \bar{k})) \in \Psi, t \in T \quad (22)$$

Constraints (23)–(25) represent the logical link for the network flow with restoration decisions. The flow through link $(i, j) \in L^k$ in network $k \in K$ is determined by the capacity of the link as well as: (i) the operational status of the nodes at both ends on that link, as shown in constraints (23) and (24), and (ii) the operational status of the link itself, as shown in constraint (25).

$$x_{ijt}^k - u_{ij}^k \beta_{it}^k \leq 0, \quad \forall (i, j) \in L^k, i \in N^k, k \in K, t \in T \quad (23)$$

$$x_{ijt}^k - u_{ij}^k \beta_{jt}^k \leq 0, \quad \forall (i, j) \in L^k, j \in N^k, k \in K, t \in T \quad (24)$$

$$x_{ijt}^k - u_{ij}^k \alpha_{ijt}^k \leq 0, \quad \forall (i, j) \in L^k, k \in K, t \in T \quad (25)$$

Finally, constraints (26)–(33) represent the nature of the decision variables in the restoration model. The amount unmet demand (slack), s_{it}^k , and flow through link $(i, j) \in L^k$, x_{ijt}^k , for infrastructure network $k \in K$ at time $t \in T$ must be non-negative, as shown in constraints (26) and (27), respectively. Constraints (28) and (29) represent the restoration status of node $i \in N^k$ and link $(i, j) \in L^k$, respectively in network $k \in K$. Likewise, the operational status of node $i \in N^k$ and link $(i, j) \in L^k$ in network $k \in K$ at time period $t \in T$ is represented by constraints (30) and (31), respectively. Constraints (32) and (33) represent the binary restoration variables for node $i \in N^k$ and link $(i, j) \in L^k$, respectively in network $k \in K$ at time $t \in T$.

$$s_{it}^k \geq 0, \quad \forall i \in N^k, k \in K, t \in T \quad (26)$$

$$x_{ijt}^k \geq 0, \quad \forall (i, j) \in L^k, k \in K, t \in T \quad (27)$$

$$z_i^k \in \{0, 1\}, \quad \forall i \in N^k, k \in K \quad (28)$$

$$y_{ij}^k \in \{0, 1\}, \quad \forall (i, j) \in L^k, k \in K \quad (29)$$

$$\beta_{it}^k \in \{0, 1\}, \quad \forall i \in N^k, k \in K, t \in T \quad (30)$$

$$\alpha_{ijt}^k \in \{0, 1\}, \quad \forall (i, j) \in L^k, k \in K, t \in T \quad (31)$$

$$\gamma_{it}^{kr} \in \{0, 1\}, \quad \forall i \in N^k, k \in K, t \in T, r \in R^k \quad (32)$$

$$\delta_{ijl}^{kr} \in \{0, 1\}, \quad \forall (i, j) \in L^k, k \in K, t \in T, r \in R^k \quad (33)$$

From the definition of the decision variable in the proposed

restoration optimization model, the model has $\sum_{k \in K} [N_d^k T + N^k(1 + T) + N^k R^k T + L^k [1 + T(1 + R^k)] + L^k T]$ variables. As for the number of constraints in the restoration model, network flow constraints (i.e., constraints (9)–(12)) have in total $\sum_{k \in K} [N^k T + L^k T]$ constraints. Constraints (13)–(21) represent the restoration decisions constraints, which consist of $\sum_{k \in K} [(N^k + L^k)(3 + T) + R^k T]$ constraints. The number of interdependent constraints (i.e., constraints (22)) is ΨT . Finally, the logical link between the network flow and the restoration decisions (i.e., constraints (23)–(25)) includes $\sum_{k \in K} [2N^k L^k T + L^k T]$. Accordingly, the total number of constraints in the proposed restoration model is $\sum_{k \in K} [N^k(3 + T) + L^k(3 + 2T) + N^k T + L^k T + 2N^k L^k T + R^k T] + \Psi T$.

The proposed optimization model has multiple objectives which could be difficult to solve since many tradeoff solutions between the multiple objectives must be identified for consideration in the restoration of a system of interdependent infrastructure networks. Hence, different multi-objective optimization techniques can be applied to find tradeoff solutions. In this paper, we use ϵ -constraint method proposed by Haimes et al. [27] to generate Pareto-optimal solutions for our restoration model as it does not aggregate the multiple objectives but instead minimizes one of them while the remaining objectives are constrained within given target values specified by decision makers. Accordingly, the multiple objectives of our proposed model, (7) and (8), can be substituted by the new objective function (34) and the additional constraint (35), where $\epsilon \in [0, 1]$ since we are dealing with the resilience of the system of interdependent infrastructure networks, and the value of the network resilience, $\mathcal{R}(t)$, at time t is between 0 and 1 (see Section 2.2).

$$\min \sum_{k \in K} \sum_{t \in T} \left(\sum_{i \in N^k} f n_{it}^k z_i^k + \sum_{(i,j) \in L^k} f l_{ijt}^k y_{ij}^k + \sum_{(i,j) \in L^k} c_{ij}^k x_{ijt}^k + \sum_{i \in N_d^k} p_{it}^k s_{it}^k \right) \quad (34)$$

$$\sum_{k \in K} \mu^k \left[\frac{\sum_{t=1}^T \left[t \left(S_d^k - \sum_{i \in N_d^k} s_{it}^k \right) - (t-1) \left(S_d^k - \sum_{i \in N_d^k} s_{it(t-1)}^k \right) \right]}{\tau (S_d^k - S_o^k)} \right] \geq \epsilon \quad (35)$$

4. Illustrative example

In this section, we illustrate our proposed restoration model with some a generated system of two interdependent infrastructure networks.

4.1. Data

Since data for real infrastructure networks are hard to obtain [5, 34], we illustrate our proposed restoration model in this paper with fictional interdependent infrastructure networks. We generate these networks using the extended algorithm for proximal topology generator proposed by Xin-Jian [65] which was originally introduced by Casey [14]. The generation process is executed in two phases: (i) generation of individual networks and (ii) building the interdependencies across them [50, 68].

In this paper, we illustrate our restoration model considering two infrastructure networks, namely simulated power and water networks, where the water network depends on the power network for operation and the power network depends on the water network for cooling and emission control [19, 68]. For the power network, the power generators are the source nodes, substations are the demand nodes, and the lines between these nodes are the links. For the water network, the water pumps are the source nodes, storage tanks are the demand nodes, and the pipelines between these nodes are the links.

For the first phase of the generation process of the interdependent infrastructure networks, each network will initially be seeded with independent and randomly distributed source nodes (i.e., no links between them). At each time step, a new randomly distributed node is added to the network and connected to nearest existing node based on Euclidean distance by adding a new undirected link between them. Then, a sparse random graph is added after the final time step to the generated network.

In the proposed restoration model, the physical dependence between the infrastructure networks is considered to describe their interdependence, that is the functionality of a node in one network is dependent on the functionality of a node in another network. Accordingly, we assume that the water pumps and storage tanks in the water network need power in order to be functional. Hence, for the second phase of the generation process of the interdependent infrastructure networks, each water pump and storage tank in the water network is depends on the nearest power supply node in the power network (i.e., power generators or substation), based on Euclidean distance.

Hence, following the above algorithm adapted from Zhang et al. [68], the two independent infrastructure networks are generated as illustrated in Fig. 2. The general properties for each network are shown in Table 1 which includes number of: nodes, undirected links, source nodes, and demand nodes, as well as the average node degree for each network.

4.2. Experiment

Considering the different possible scenarios of disruptions discussed in Section 2.3 (“Random” for random failures, “Capacity” for capacity-based malevolent attacks, “Degree” for degree-based malevolent attacks, and “Spatial” for spatial failures), the efficiency of each interdependent infrastructure network with the removal of a fraction of

Table 1

General properties of the interdependent networks.

Network	N	L	N^s	N^d	deg
Power	25	31	5	5	2.48
Water	25	35	5	5	2.8

components (nodes or links) of the system of interdependent networks is illustrated in Fig. 3. Efficiency is measured as the ratio of the current max flow over the original max flow. It can be observed from Fig. 3 that the removal of the components of the system of interdependent infrastructure networks with the highest capacity result in the largest decline in the individual network efficiencies as the fraction of components removal increases. On the other hand, the spatial removal of the components of the system of interdependent infrastructure networks mostly result in the smallest drop in the individual networks efficiencies among other disruptions scenarios which could because of the existence of alternative routes within the networks since it affects a specific area in the network.

To assess the proposed multi-objective restoration model, a subset of the Pareto optimal set (i.e. non-dominated solutions) were obtained using LINGO 17.0 by varying the value of ϵ and solve the optimization model again for each value of ϵ . Fig. 4 illustrates the generation of different points of the subset of Pareto front using different values of ϵ (i.e., $\epsilon \in [0.5, 1]$) considering the availability of one work crew for each network during the restoration process with different possible scenarios of disruptions, see Section 2.3. In addition, the restoration cost is considered for Fig. 4 to be higher than the unmet demand cost for node $i \in N^k$ in network $k \in K$ (i.e., $fn_{it}^k > p_{it}^k$). Otherwise, the resilience will always be 1 given that there is enough time to recover the essential components because both objectives are focused on unmet demand, and by doing so, the resilience of the system of interdependent networks

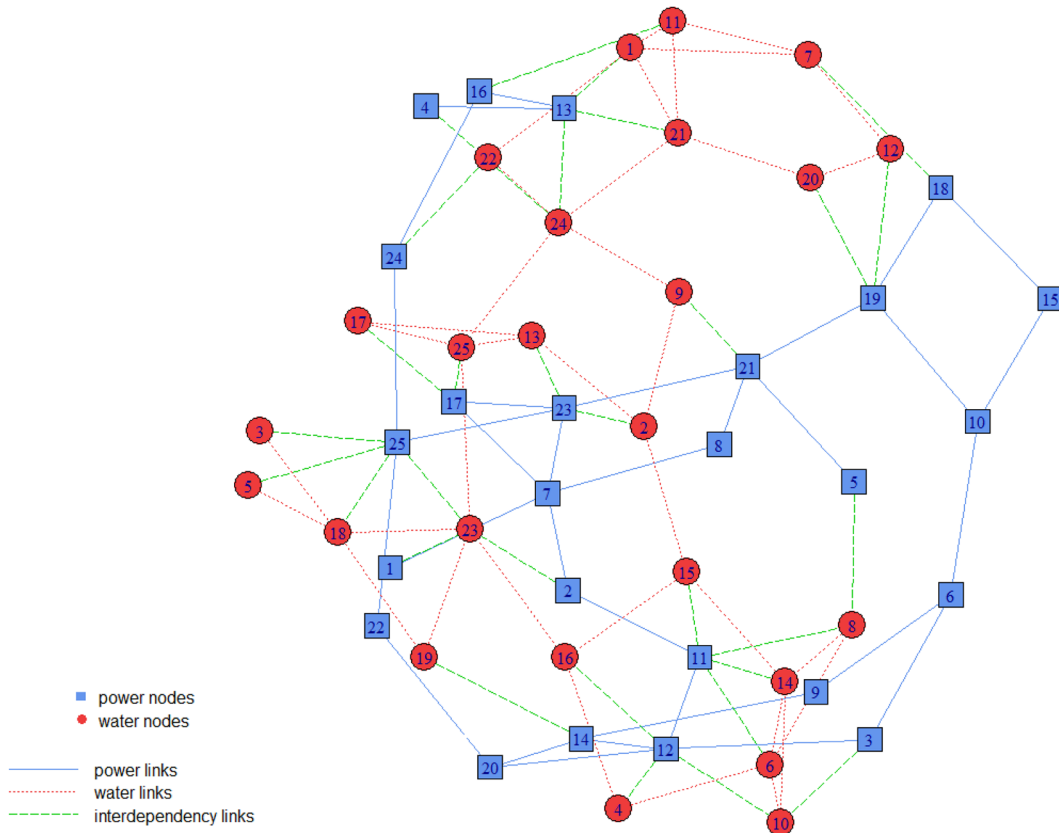


Fig. 2. An interdependent network example.

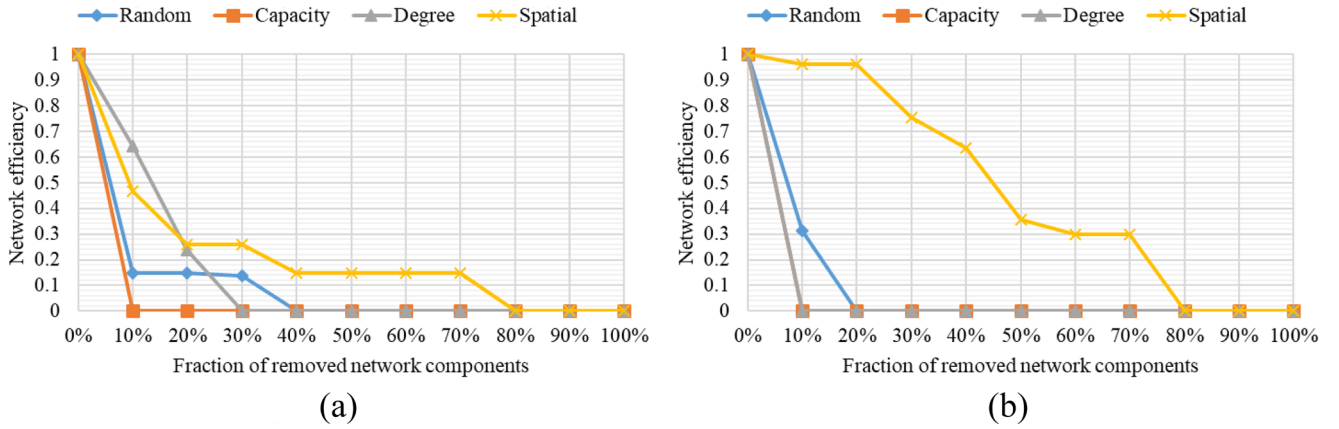


Fig. 3. Network efficiency with fraction components removals of the interdependent networks considering four disruptions scenarios for the (a) power network, and (b) water network.

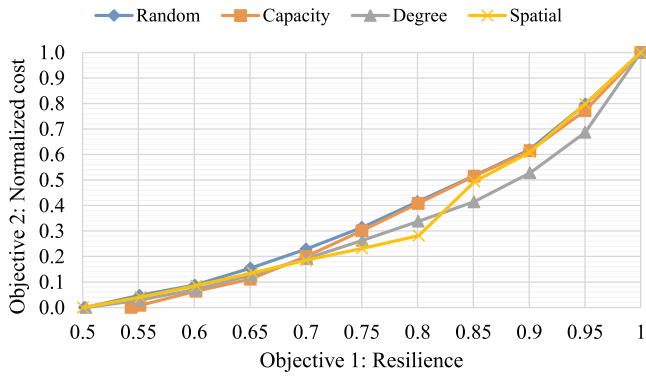


Fig. 4. Objectives tradeoffs.

will be maximized and the total cost associated with the restoration will be minimized as well. In Fig. 4, the horizontal axis represents the resilience of the system of interdependent infrastructure networks for which a maximum is sought, while the vertical axis represents total cost (i.e., restoration cost, flow cost, and disruption cost) that we would like to minimize. As observed in Fig. 4, the total cost associated with the restoration process increases as the value of ε (i.e., the minimum value of the interdependent networks resilience desired) increases. The lowest value of objective 1 (i.e. resilience) for all different scenarios of disruptions in Fig. 4 is when $\varepsilon = 0.5$ while the highest value is when $\varepsilon = 1$. Hence, Fig. 4 serves to illustrate the tradeoffs between the two objectives of the restoration optimization model where when a higher level

of resilience is desired for the system of interdependent networks, the total cost associated with the restoration process will be higher.

Without loss of generality, we consider the following parameter distributions and values for illustrative purposes of the proposed multi-objective restoration model: $\mu^k = 1/|K|$, $\tau = 50$, $fn_{it}^k, fl_{ij}^k, u_{ij}^k \sim U(20, 50)$, $c_{ij}^k \sim U(1, 10)$, $p_{it}^k = 60$, and $dn_{it}^k, dl_{ij}^k \sim U(1, 5)$. In this example, though the restoration cost of disrupted nodes and links are different (i.e., not the same for all disrupted components), they are assumed to be the same for all time periods. That is, the restoration time for a disrupted component does not change with time. Similarly, the cost of unmet demand in this example is assumed to be the same for all time periods. In this work, The disruption cost for node $i \in N^k$ in network $k \in K$ (i.e., unmet demand cost, p_{it}^k) is considered higher than its restoration cost, fn_{it}^k , since we are aiming to maximize the resilience of the system of interdependent infrastructure networks. Hence, both objectives will be focusing on minimizing the unmet demand at node $i \in N^k$ in network $k \in K$. Figs. 5 through 8 depict the trajectory of system of interdependent infrastructure network resilience considering the three different scenarios for availability of the work crews (WC): one work crew ("1 WC"), two work crews ("2 WCs"), and three work crews ("3 WCs"), with random, capacity-based, degree-based, and spatial disruption scenarios, respectively which could help decision makers when developing their restoration plans following a disruptive event (e.g., considering more work crews for one network than the other). The percentages of disrupted components in the system of interdependent power-water networks are: 20.9%, 20.9%, 20.9%, and 21.4% for the scenarios of random, capacity-based, degree-based, and spatial disruptions,

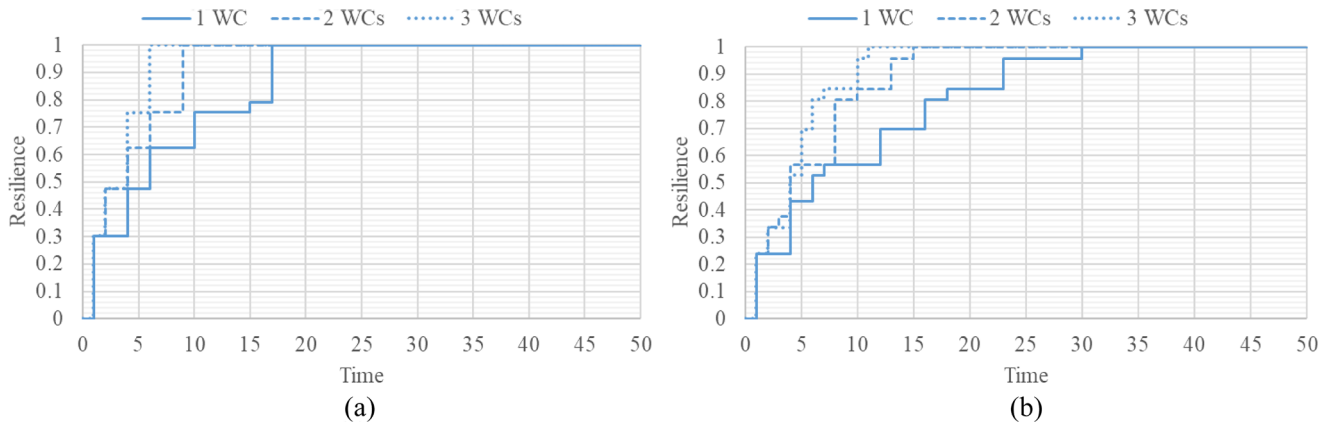


Fig. 5. Network resilience over the recovery time horizon for a random disruption with different number of work crews for the (a) power network, and (b) water network.

respectively. The disrupted components in the first three disruptions scenarios are: 10 nodes, 5 from each network, and 14 bi-directional links, 7 from each network. For the spatial disruptions: 11 nodes, 5 from the power network and 6 from the water network, and 14 bi-directional links, 7 from the power network and 7 from the water network. Moreover, the number of removed components from the system of interdependent infrastructure networks in the first three disruption scenarios are equal, 10 nodes and 14 links, as we are removing them individually according to specific criteria (i.e., random, highest capacity, highest degree). However, for the spatial disruption scenario, they are removed according to their locations (i.e., if a disruption occurs in a region, all the components of the system of interdependent networks within that region will be disrupted and hence removed from their networks).

As can be observed from Figs. 5 through 8, considering more work crews helps in achieving full resilience of the system of interdependent infrastructure networks earlier. That is, increasing the number of work crews in each network could reduce the time to full resilience of the system of interdependent infrastructure networks. However, it should be noted that it could take one network longer time reach a full resilience status (i.e., retain its performance prior to the occurrence of a disruptive event) than other networks within the same system of interdependent networks even if they are assigned same number of work crews. This could result due to one of the following possible reasons: (i) the nature of interdependencies among such networks within the system, (ii) the restoration durations of the disrupted components in each network in the system, or (iii) both (i) and (ii). For example, water network took longer time to be fully resilient, considering four different disruption scenarios, than power network as it depends on some nodes on power network that were disrupted and need to be restored first, as shown in Figs. 5 through 8. Similarly, the power network could also take a longer time to regain its performance prior to a disruption if it depends on some disrupted nodes in the water network or its disrupted components have higher restoration durations than the ones for disrupted components in water. Hence, assigning more work crews to restore the disrupted components in one network than the other could help in expediting the restoration process of that network, accordingly reaching the maximum level of resilience of the system of interdependent infrastructure networks faster considering the available time periods. In general, there are three factors that affect the progress of improvement for the resilience of the system of interdependent infrastructure networks: (i) the set of disrupted components in the system of interdependent networks, (ii) the nature of the interdependencies among the infrastructure networks, and (iii) the number of available work crews for each infrastructure network during the restoration process. Furthermore, the available time horizon and budget for the restoration process can decide what will be the maximum level of

resilience that the system of interdependent infrastructure networks can reach. Accordingly, what are the disrupted components in the system of interdependent networks that need to be restored. Fig. 6, Fig. 7 and Fig. 8.

The set of disrupted components (“DC”), nodes and links, in power and water networks considering four different disruption scenarios, discussed earlier in Section 2.3, are shown in Tables 2 and 3, respectively. Tables 2 and 3 provide the combination of the restoration time for each disrupted component in power and water networks, respectively that is selected to be restored (i.e., the time at which a disrupted component is restored by one of the available work crews) and the work crew who restored that disrupted components (“RT,WC”) considering three different scenarios for availability of the work crews (WC): one work crew (“1 WC”), two work crews (“2 WCs”), and three work crews (“3 WCs”) with four different disruption scenarios, random, capacity-based, degree-based, and spatial disruption scenarios. For example, considering the availability of two worker crews (2 WCs) with random disruption scenario for power network, node 23 (i.e., DC = 23) is restored at time 6 and assigned to WC 1 (i.e., RT,WC = 6,1) and link (6,10) (i.e., DC = (6,10)) is restored at time 4 and assigned to WC 2 (i.e., RT,WC = 4,2), as illustrated in Table 2. Similarly, considering the availability of three work crews (3 WCs) with spatial disruption scenario for water network, node 19 (i.e., DC = 19) is restored at time 10 and assigned to work crew 3 (i.e., RT,WC = 10,3) and link (16,23) (i.e., DC = (16,23)) is restored at time 2 and assigned to work crew 1 (i.e., RT,WC = 2,1), as shown in Table 3.

Moreover, there could be a disrupted network component in any network that is not selected to be restored due to one of two reasons: (i) this disrupted component is not influential for the resilience of the system of interdependent networks (i.e., restoring such component will not enhance the resilience of the system of interdependent networks), or (ii) restoring this disrupted component costs more than what that restoration would save in the flow cost. Accordingly, such network component is not restored (“NR”) as shown in Tables 2 and 3 for power and water networks, respectively. For example, four components were not restored in power network considering degree-based disruption scenario, see Table 2. Likewise, five components were not restored in water network considering degree-based disruption scenario, as shown in Table 3. Table 3 shows that all disrupted components in water network are restored when considering the spatial disruption scenario, which mean all of the disrupted component are critical for the system of interdependent network (i.e., restoring these disrupted component will enhance the resilience of the system).

The proposed resilience-driven multi-objective restoration model focuses on maximizing the resilience of the system of interdependent infrastructure networks to retain their performance level prior to the disruption. Hence, the disrupted components might not be all restored,

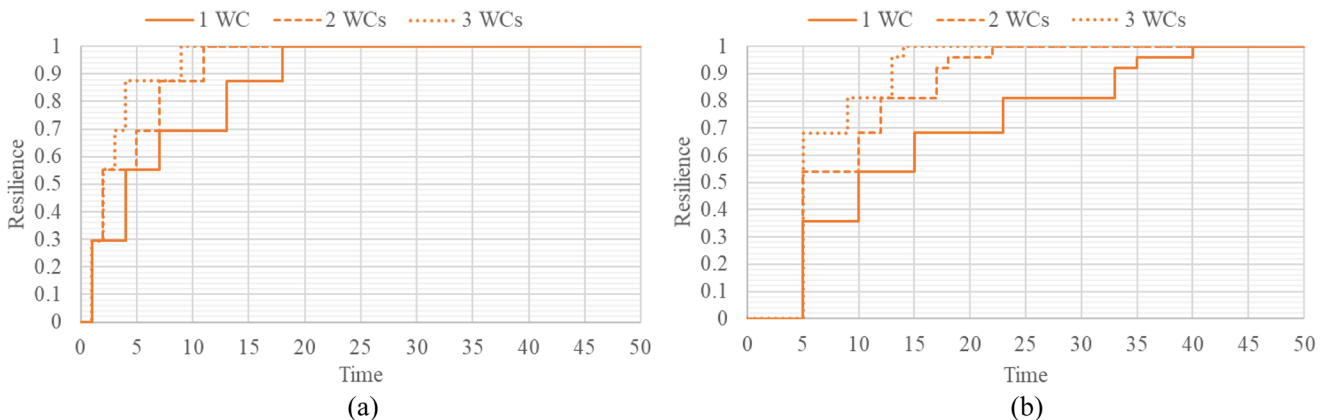


Fig. 6. Network resilience over the recovery time horizon for a capacity-based disruption with different number of work crews for the (a) power network, and (b) water network.

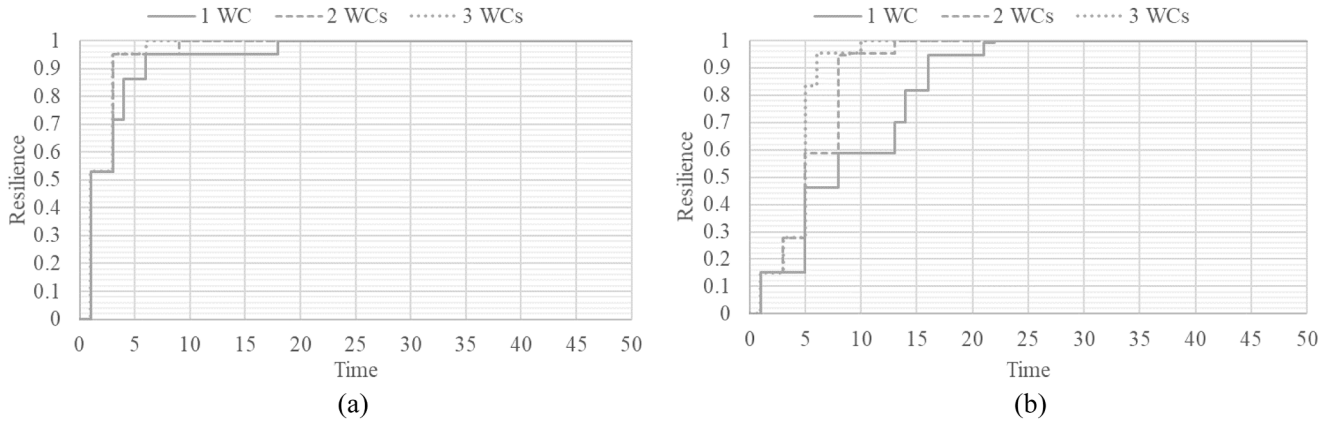


Fig. 7. Network resilience over the recovery time horizon for a degree-based disruption with different number of work crews for the (a) power network, and (b) water network.

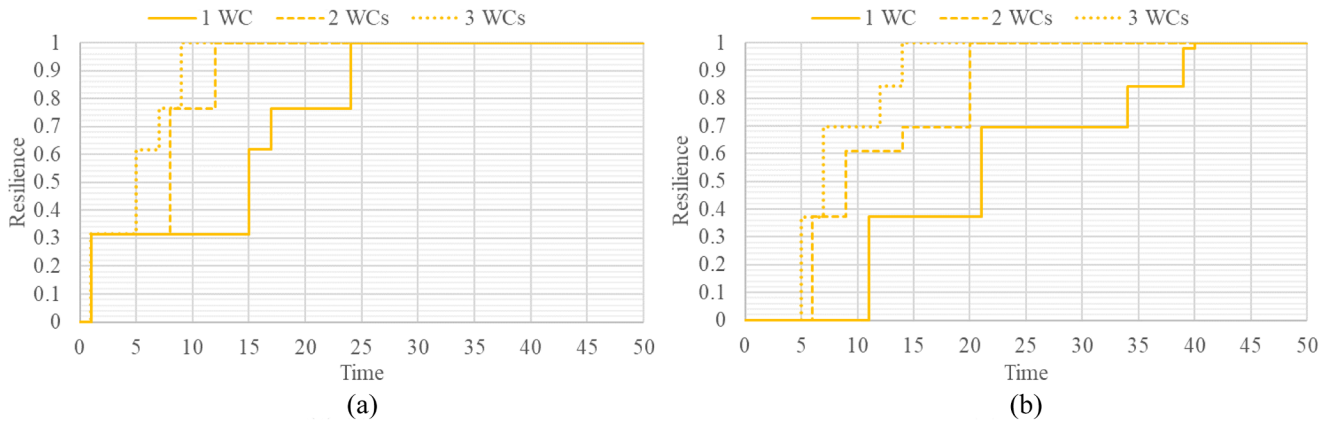


Fig. 8. Network resilience over the recovery time horizon for a spatial disruption with different number of work crews for the (a) power network, and (b) water network.

Table 2

Restoration time and assignment for disrupted components in the power network.

Random DC	RT,WC			Capacity-based DC	RT,WC			Degree-based DC	RT,WC			Spatial DC	RT,WC		
	1 WC	2 WCs	3 WCs		1 WC	2 WCs	3 WCs		1 WC	2 WCs	3 WCs		1 WC	2 WCs	3 WCs
7	10,1	5,1	4,1	3	18,1	11,2	9,3	7	3,1	3,1	3,1	1	4,1	8,2	5,3
13	4,1	2,2	2,2	7	7,1	5,1	3,3	11	NR	NR	NR	2	24,1	11,1	9,3
21	1,1	1,1	1,1	21	4,1	2,1	2,1	12	12,1	9,1	6,1	7	10,1	6,2	4,2
23	7,1	6,1	5,2	23	11,1	6,2	4,3	23	4,1	1,2	1,3	22	1,1	1,1	1,2
35	NR	NR	NR	25	10,1	5,2	5,1	25	9,1	6,1	5,2	25	7,1	3,2	3,3
(1,7)	15,1	9,2	6,3	(5,21)	3,1	2,2	2,2	(3,12)	18,1	9,1	6,3	(1,7)	15,1	8,1	5,1
(2,11)	NR	NR	NR	(7,23)	13,1	7,1	4,2	(7,23)	6,1	3,2	2,2	(2,7)	21,1	12,2	9,2
(6,10)	6,1	4,2	4,2	(12,14)	NR	NR	NR	(11,12)	NR	NR	NR	(2,11)	NR	NR	NR
(7,23)	17,1	8,1	6,1	(13,16)	1,1	1,1	1,1	(12,14)	NR	NR	NR	(7,8)	2,1	12,1	5,2
(10,19)	21,1	12,1	9,2	(17,23)	NR	NR	NR	(12,20)	13,1	4,2	6,1	(7,17)	NR	NR	NR
(13,16)	2,1	2,1	1,3	(18,19)	20,1	9,1	8,2	(21,23)	NR	NR	NR	(7,23)	17,1	3,1	7,1
(17,23)	NR	NR	NR	(21,23)	NR	NR	NR	(23,25)	NR	NR	NR	(22,25)	NR	NR	NR

especially if they do not have an effect on the resilience of the other networks. Accordingly, the full resilience of the system of interdependent infrastructure networks could be achieved prior to complete restoration of these networks (i.e., time to full resilience (TFR) $\leq$ time to complete restoration (TCR) [7,9]). Table 4 shows a comparison between the time when the system of interdependent power-water networks is fully resilient and the time when all the disrupted components of the system of interdependent networks are restored considering the different disruptions scenarios, discussed earlier in Section 2.3, along with three different scenarios for availability of the work crews (i.e., one, two, or three work crews).

As shown in Table 4, the time to full resilience for the system of interdependent networks is less than the time to complete restoration of the all disrupted components in the system when considering the disruption scenarios of random, capacity-based, and degree-based with all three levels of work crews availability. This indicates that not all the disrupted components are influential for the resilience of the system of interdependent networks; hence they are not required to be restored. On the other hand, considering the spatial disruption scenario, the time to full resilience for the system of interdependent networks equals the time to complete restoration of the all disrupted components in the system. This could be a result of one of two reasons: (i) all the disrupted

Table 3

Restoration time and assignment for disrupted components in the water network.

Random DC	RT,WC			Capacity-based DC	RT,WC			Degree-based DC	RT,WC			Spatial DC	RT,WC		
	1 WC	2 WCs	3 WCs		1 WC	2 WCs	3 WCs		1 WC	2 WCs	3 WCs		1 WC	2 WCs	3 WCs
2	7,1	5,1	5,3	1	5,1	5,1	5,1	1	5,1	5,2	5,3	3	15,1	12,2	7,3
6	NR	NR	NR	2	20,1	11,1	6,1	6	NR	NR	NR	5	35,1	7,2	13,2
9	12,1	7,2	5,1	23	10,1	5,2	5,2	14	NR	NR	NR	16	4,1	6,2	4,3
13	16,1	9,1	6,2	24	23,1	8,2	8,3	23	13,1	8,2	5,2	18	12,1	9,2	7,2
21	4,1	4,1	4,3	25	33,1	16,1	13,3	24	8,1	3,2	3,2	19	34,1	15,2	10,3
(6,10)	NR	NR	NR	(2,9)	19,1	12,2	9,2	(1,21)	21,1	13,2	10,2	23	9,1	5,1	5,2
(10,14)	NR	NR	NR	(2,13)	28,1	17,2	11,1	(6,14)	NR	NR	NR	(3,18)	21,1	14,1	7,1
(12,20)	25,1	16,2	11,3	(2,15)	41,1	21,1	14,3	(16,23)	16,1	8,1	5,1	(5,18)	39,1	9,1	14,3
(13,25)	18,1	11,1	7,3	(3,18)	15,1	10,1	5,3	(18,23)	14,1	6,1	6,1	(16,23)	11,1	2,2	2,1
(16,23)	6,1	2,2	2,2	(6,8)	NR	NR	NR	(21,24)	NR	NR	NR	(18,19)	26,1	19,1	12,2
(19,23)	23,1	12,2	10,1	(13,25)	35,1	18,1	13,1	(23,25)	22,1	9,1	6,3	(18,23)	16,1	8,2	6,2
(20,21)	30,1	16,1	11,2	(15,16)	40,1	22,2	14,2	(24,25)	NR	NR	NR	(19,23)	31,1	20,2	12,1
												(23,25)	40,1	20,1	14,2

Table 4

Comparison between time to full resilience (TFR) and time to complete restoration (TCR) for the system of interdependent networks considering different disruption scenarios with different number of work crews.

Disruption Scenario	One work crew		Two work crews		Three work crews	
	TFR	TCR	TFR	TCR	TFR	TCR
Random	30	45	21	31	11	16
Capacity	41	50	26	31	14	18
Degree	22	41	16	29	10	20
Spatial	40	40	20	20	14	14

components are influential for the resilience of the system of interdependent networks, accordingly all of them need to be restored to retain the performance of the system prior to the disruption (this is not the case for our example since there are some disrupted components in the power network that are not restored as restoring them will not enhance the resilience of the system, as shown in Table 2); or (ii) non-critical disrupted components of one network can be restored (i.e., to get the time to full restoration of disrupted components) prior to the time at which another network in the system becomes fully resilient (this is the case for our example). For example, the three non-critical components in the power network are restored prior to the time at which the water network becomes fully resilient (i.e., the time needed to restore all the critical disrupted components in the water network).

5. Concluding remarks

Infrastructure networks are increasingly becoming more interdependent, potentially making them highly vulnerable. These interdependent infrastructure networks can be disrupted in different scenarios (i.e., random failures, malevolent attacks, or spatial disruptions). Consequently, a disruption in one infrastructure network could lead to disruptions in other infrastructure networks due to their interdependencies. Therefore, the restoration process of these interdependent infrastructure networks following any disruptive event has become a challenging task for decision makers.

In this paper, we study the restoration problem of a system of interdependent infrastructure networks given the occurrence of a disruptive event considering different disruptions scenarios and the physical interdependence between the infrastructure networks. We propose a multi-objective resilience-driven restoration model using mixed-integer programming with the objective of maximizing the resilience of the system of interdependent networks over the restoration time horizon while the total cost as the sum of flow cost, restoration cost, and disruption cost (i.e., unmet demand cost) is minimized. The model takes into account the availability of the time periods along with the resources, as well as the different restoration times for the disrupted

components. It provides the set of disrupted components to be restored in order to have a fully resilient system of interdependent networks and allocate and schedule these disrupted components to the available work crews, noting that not necessarily all disrupted components need to be restored to achieve full resilience. The proposed model is illustrated through a generated system of interdependent power-water networks, though the approach is applicable to any system of physically interdependent networks. Managerial insights for utility companies and municipalities come in the form of determining an optimal restoration plan for infrastructure network components and a schedule to assign work crews to those restoration tasks. Managers can determine, for example, the number of work crews by comparing recovery trajectories in various networks or locations where equipment can be stored to expedite restoration tasks [43]. Such insights also highlight the benefits of centralized decision making, which assumes that all utilities co-operate for a common objective.

The proposed restoration model assigns one work crew to restore a disrupted component which could be extended to allow the assignment of more than one work crew to the same disrupted component if it is so critical in order to restore it at an earlier time as some components have a huge impact on the efficiency of their network. In addition, the proposed restoration model considers binary disruption status for the disrupted components in the interdependent networks (i.e., a network component is either fully disrupted or undisrupted). Hence, a partial disruption status could be considered for the disrupted networks component which could allow for partial functioning of the disrupted components and dependent components as well. Moreover, studying the vulnerability of the components in each infrastructure network could help in identifying the critical ones to reinforce or protect prior to any disruption, thus potentially leading a shorter time to achieve full resilience as well as a lower cost associated with the restoration process. Therefore, a tradeoff between the vulnerability and restoration of the system of interdependent infrastructure networks could be studied to find the optimal strategy for investment. Furthermore, heuristic or metaheuristics method could be tried to solve the proposed restoration model for larger problems in a timely manner. Also, the proposed model considers only the physical interdependency among infrastructure networks. However, other types of interdependency could be considered such as geographical interdependency. Finally, the proposed restoration model could be extended to consider network-specific constraints and apply the new restoration model on a real-life system of interdependent infrastructure networks.

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